

Name: \_\_\_\_\_

# VOLUME AND SURFACE AREA, (SURFACE AREA OF PRISMS)

Date: \_\_\_\_\_

**LO: I can calculate the surface area of prisms by finding the area of each face.**

## Surface area of prisms

A **prism** is a 3D shape with a uniform cross-section. Its **surface area** is the sum of the areas of all its faces: two identical end faces plus the rectangular side faces.

**Method:** Find the area of the cross-section (end face), double it for both ends, then add the areas of each rectangular side.  $SA = 2 \times (\text{area of cross-section}) + (\text{perimeter of cross-section}) \times \text{length}$ .

## Quick examples

|   |                                                          |                                     |
|---|----------------------------------------------------------|-------------------------------------|
| ● | Cuboid: $SA = 2(lw + lh + wh)$                           | three pairs of rectangular faces    |
| ● | Triangular prism: 2 triangles + 3 rectangles             | count all faces                     |
| ● | $SA = 2A + Ph$ (A = end area, P = perimeter, h = length) | general prism formula               |
| ● | $SA = \text{area of cross-section} \times \text{length}$ | that gives volume, not surface area |

## Key idea

Surface area of a prism = 2 times the end face area plus the lateral (side) area. The lateral area equals perimeter of cross-section times length.

$$SA = 2A + P \times l$$

A = area of end face, P = perimeter of end, l = length of prism

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - cuboid

Find the surface area of a cuboid 6 cm by 4 cm by 3 cm.

Three pairs of faces:  $6 \times 4 = 24$ ,  $6 \times 3 = 18$ ,  $4 \times 3 = 12$

Double each:  $48 + 36 + 24$

$$SA = 108 \text{ cm}^2$$

$$108 \text{ cm}^2$$

A cuboid has three different pairs of rectangular faces.

### Example 2 - triangular prism

A triangular prism has a right-angled triangle cross-section (base 6, height 8, hypotenuse 10 cm) and length 12 cm. Find the SA.

$$\text{Two triangles: } 2 \times \frac{1}{2} \times 6 \times 8 = 48 \text{ cm}^2$$

$$\text{Three rectangles: } 6 \times 12 + 8 \times 12 + 10 \times 12 = 72 + 96 + 120 = 288 \text{ cm}^2$$

$$SA = 48 + 288 = 336 \text{ cm}^2$$

$$336 \text{ cm}^2$$

Find all five faces: two triangles and three rectangles.

### Example 3 - L-shaped prism

An L-shaped prism has cross-section area  $20 \text{ cm}^2$ , perimeter 24 cm, and length 10 cm. Find the SA.

$$\text{Two ends: } 2 \times 20 = 40 \text{ cm}^2$$

$$\text{Lateral area: } 24 \times 10 = 240 \text{ cm}^2$$

$$SA = 40 + 240 = 280 \text{ cm}^2$$

$$280 \text{ cm}^2$$

Use the general formula when end area and perimeter are given.

### Example 4 - trapezoidal prism

A prism has a trapezium cross-section with parallel sides 5 cm and 9 cm, height 4 cm, slant sides 5 cm each, and length 15 cm.

$$\text{End area: } \frac{1}{2}(5 + 9) \times 4 = 28 \text{ cm}^2. \text{ Two ends} = 56 \text{ cm}^2$$

$$\text{Perimeter: } 5 + 9 + 5 + 5 = 24 \text{ cm}$$

$$\text{Lateral area: } 24 \times 15 = 360 \text{ cm}^2. SA = 56 + 360 = 416 \text{ cm}^2$$

$$416 \text{ cm}^2$$

Find the trapezium area, then use  $SA = 2A + Pl$ .

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Cuboids

Find the surface area of each cuboid.

- a)  $5\text{ cm} \times 3\text{ cm} \times 2\text{ cm}$
- b)  $8\text{ cm} \times 4\text{ cm} \times 3\text{ cm}$
- c)  $10\text{ cm} \times 6\text{ cm} \times 4\text{ cm}$
- d)  $7\text{ cm} \times 7\text{ cm} \times 7\text{ cm}$

### 2. Triangular prisms

Find the surface area of each triangular prism.

- a) Triangle: base 6, height 4, hyp 5 cm.  
Length 10 cm.
- b) Triangle: base 8, height 3, hyp 5 cm.  
Length 12 cm.
- c) Triangle: base 5, height 12, hyp 13 cm.  
Length 8 cm.
- d) Equilateral triangle side 6 cm. Length 10 cm.

### 3. Using $SA = 2A + Pl$

Given the cross-section area A, perimeter P, and length l, find the SA.

- a)  $A = 15\text{ cm}^2$ ,  $P = 20\text{ cm}$ ,  $l = 8\text{ cm}$
- b)  $A = 24\text{ cm}^2$ ,  $P = 22\text{ cm}$ ,  $l = 10\text{ cm}$
- c)  $A = 30\text{ cm}^2$ ,  $P = 28\text{ cm}$ ,  $l = 6\text{ cm}$
- d)  $A = 18\text{ cm}^2$ ,  $P = 18\text{ cm}$ ,  $l = 12\text{ cm}$

### 4. Trapezoidal prisms

Find the surface area of each trapezoidal prism.

- a) Parallel sides 4, 8 cm; height 3 cm; slant sides 5 cm each; length 10 cm
- b) Parallel sides 6, 10 cm; height 4 cm; slant sides 5 cm each; length 12 cm
- c) Parallel sides 3, 7 cm; height 3 cm; slant sides 4 cm each; length 9 cm
- d) Parallel sides 5, 11 cm; height 4 cm; slant sides 5 cm each; length 14 cm

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) SA of a prism =  $2 \times \text{end area} + \text{perimeter} \times \text{length}$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) SA of a prism =  $\text{cross-section area} \times \text{length}$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) A triangular prism has 5 faces ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) All faces of a prism are rectangles ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) To find SA you must add the area of every single face ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A chocolate box is a triangular prism with an equilateral triangle cross-section of side **8 cm** and length **20 cm**. The box needs wrapping with **10% extra** paper for overlaps. How much wrapping paper is needed in  $\text{cm}^2$ ?

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) A swimming pool has a trapezoidal cross-section: one end is **1 m deep**, the other is **2.5 m deep**, the pool is **25 m long** and **10 m wide**. Find the total internal surface area that needs tiling (base and all four sides, no top).

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

